NEET (UG) 2027 — Expected Question Paper

Physics · Chemistry · Biology (Botany & Zoology) | 180 questions · 720 marks · 3 hours

Instructions

1. All 180 questions are compulsory: Physics 1–45, Chemistry 46–90, Botany 91–135, Zoology 136–180.
2. Each correct answer earns 4 marks; each wrong answer loses 1 mark; unanswered questions score 0.
3. Choose the single best option (1), (2), (3) or (4).
4. The answer key and explanations are at the end. Attempt the full paper in 3 hours before checking.

How this paper was built

This is an original practice paper, not a leaked or official paper. It follows the NEET 2025 pattern (180 compulsory questions, no optional section). The chapter weightage mirrors the most frequently tested NCERT topics in NEET 2015–2025, the same papers collected in the NEET folder. It is weighted towards high-yield chapters: Mechanics, Electrostatics, Optics and Modern Physics; Bonding, Equilibrium and Coordination Compounds; Genetics, Plant Physiology, Human Physiology, Reproduction and Biotechnology.

Chapter-wise distribution

Physics (45)	
Units & Dimensions	1
Kinematics	2
Laws of Motion	2
Work, Energy & Power	3
Rotational Motion	3
Gravitation	2
Mechanical Properties of Solids	1
Mechanical Properties of Fluids	3
Thermodynamics	1
Kinetic Theory	2
Oscillations	2
Waves	2
Electrostatics	2
Capacitance	2
Current Electricity	3
Moving Charges & Magnetism	3
Electromagnetic Induction	1
Alternating Current	2
Electromagnetic Waves	1
Ray Optics	1
Wave Optics	1
Dual Nature	2
Atoms & Nuclei	2
Semiconductors	1
Chemistry (45)	
Some Basic Concepts	2
Atomic Structure	1
Classification of Elements	2
Chemical Bonding	4

Thermodynamics	1
Equilibrium	3
Solutions	1
Electrochemistry	2
Chemical Kinetics	2
p-Block	5
d- and f-Block	4
Coordination Compounds	3
Organic: Basic Principles	3
Hydrocarbons	3
Haloalkanes	1
Alcohols & Phenols	1
Aldehydes & Ketones	2
Amines	2
Biomolecules	3
Botany (45)	
Biological Classification	2
Plant Kingdom	3
Morphology of Flowering Plants	3
Anatomy of Flowering Plants	3
Cell: Unit of Life	2
Cell Cycle	2
Biomolecules	1
Transport in Plants	2
Mineral Nutrition	1
Photosynthesis	3
Respiration	2
Plant Growth	3
Sexual Reproduction in Plants	4
Principles of Inheritance	4
Molecular Basis of Inheritance	4
Ecosystem	2
Biodiversity	2
Organisms & Populations	2
Zoology (45)	
Animal Kingdom	4
Structural Organisation	3
Digestion	3
Breathing & Exchange of Gases	2
Body Fluids & Circulation	3
Excretion	3
Locomotion & Movement	3
Neural Control	2
Chemical Coordination	3
Human Reproduction	5
Reproductive Health	3
Evolution	3
Human Health & Disease	4
Biotechnology: Principles	2
Biotechnology: Applications	2

Physics

1. Planck's constant has the same dimensions as
(1) Energy
(2) Power
(3) Angular momentum
(4) Linear momentum

2. A ball is thrown vertically upward with a speed of 20 m s^{-1}. Taking g = 10 m s^{-2}, the maximum height reached is
(1) 80 m
(2) 20 m
(3) 40 m
(4) 10 m

3. Two projectiles are launched with the same speed at angles 30° and 60° with the horizontal. The ratio of their horizontal ranges is
(1) $\sqrt{3} : 1$
(2) $1 : \sqrt{3}$
(3) 1 : 1
(4) 1 : 3

4. A horizontal force of 20 N acts on a 5 kg block on a frictionless surface. The acceleration of the block is
(1) 5 m s^{-2}
(2) 4 m s^{-2}
(3) 2 m s^{-2}
(4) 100 m s^{-2}

5. A block rests on a rough inclined plane with coefficient of static friction 0.5. The block just begins to slide when the angle of inclination is
(1) 30°
(2) $\tan^{-1}(0.5)$
(3) $\sin^{-1}(0.5)$
(4) 45°

6. A body of mass 2 kg has kinetic energy 100 J. Its linear momentum is
(1) 200 kg m s^{-1}
(2) 20 kg m s^{-1}
(3) 40 kg m s^{-1}
(4) 10 kg m s^{-1}

7. The work done in stretching a spring of force constant 200 N m^{-1} by 10 cm from its natural length is
(1) 20 J
(2) 2 J
(3) 1 J
(4) 10 J

8. A body moving with speed v collides elastically head-on with an identical body at rest. After the collision
(1) The first body rebounds with v
(2) Both move with v/2
(3) The first body comes to rest and the second moves with v
(4) Both move together with v

9. The moment of inertia of a solid sphere of mass M and radius R about a tangent to its surface is
(1) $(2/5)MR^2$
(2) $(5/3)MR^2$
(3) $(3/2)MR^2$
(4) $(7/5)MR^2$

10. For a solid cylinder rolling without slipping, the fraction of its total kinetic energy that is rotational is
(1) 1/4
(2) 1/3
(3) 1/2
(4) 2/3

11. A disc rotating at angular speed ω has a ring of equal moment of inertia gently placed coaxially on it. The final angular speed is
(1) 2ω
(2) ω
(3) ω/2
(4) ω/4

12. The escape speed from Earth is 11.2 km s^{-1}. For a planet of the same mean density but twice the radius of Earth, the escape speed is
(1) 5.6 km s^{-1}
(2) 22.4 km s^{-1}
(3) 11.2 km s^{-1}
(4) 44.8 km s^{-1}

13. For a satellite orbiting very close to the Earth's surface, the ratio of escape speed to orbital speed is
(1) $\sqrt{2}$
(2) 2
(3) $1/\sqrt{2}$
(4) 1

14. A wire stretches by ΔL under a load. Another wire of the same material with double the length and double the radius, under the same load, stretches by
(1) ΔL/4
(2) 2ΔL
(3) ΔL/2
(4) ΔL

15. If the radius of a small spherical drop falling through a viscous liquid is doubled, its terminal velocity becomes
(1) Eight times
(2) Four times
(3) Double
(4) Half

16. Water flows at 2 m s^{-1} through a pipe of cross-section 4 cm^2. Its speed where the cross-section is 1 cm^2 is
(1) 2 m s^{-1}
(2) 0.5 m s^{-1}
(3) 4 m s^{-1}
(4) 8 m s^{-1}

17. The excess pressure inside a soap bubble of radius r and surface tension T is
(1) T/r
(2) 2T/r
(3) 8T/r
(4) 4T/r

18. A Carnot engine works between 500 K and 300 K. Its efficiency is
(1) 67%
(2) 40%
(3) 60%
(4) 20%

19. The temperature of a gas is raised from 27 °C to 327 °C. Its rms speed becomes
(1) 12 times
(2) Half
(3) 2 times
(4) √2 times

20. For a diatomic ideal gas at room temperature (no vibration), $\gamma = C_p/C_v$ is
(1) 9/7
(2) 4/3
(3) 5/3
(4) 7/5

21. The time period of the SHM $x = 5\sin(\pi t + \pi/3)$ cm (t in s) is
(1) 2 s
(2) 1 s
(3) 0.5 s
(4) π s

22. If the length of a simple pendulum is increased by 21%, its time period increases by
(1) 10.5%
(2) 11%
(3) 10%
(4) 21%

23. The wave $y = 0.02\sin[2\pi(50t - 0.5x)]$ (x in m, t in s) travels with speed
(1) 25 m s^{-1}
(2) 100 m s^{-1}
(3) 200 m s^{-1}
(4) 50 m s^{-1}

24. A closed organ pipe has fundamental frequency 100 Hz. The frequency of its first overtone is
(1) 300 Hz
(2) 400 Hz
(3) 150 Hz
(4) 200 Hz

25. The force between two point charges is F. If each charge is doubled and their separation is also doubled, the force becomes
(1) 4F
(2) F
(3) 2F
(4) F/4

26. An electric dipole is placed in a uniform electric field. It experiences
(1) A net force but no torque
(2) Neither force nor torque in any orientation
(3) A net force and a torque
(4) A torque but zero net force

27. Three capacitors of 6 μF each are connected in series. The equivalent capacitance is
(1) 2 μF
(2) 18 μF
(3) 6 μF
(4) 3 μF

28. The energy stored in a 10 μF capacitor charged to 100 V is
(1) 0.05 J
(2) 5 J
(3) 0.005 J
(4) 0.5 J

29. A wire of resistance R is stretched uniformly to double its length. Its new resistance is
(1) R/2
(2) 4R
(3) R/4
(4) 2R

30. In a balanced Wheatstone bridge P = 10 Ω, Q = 20 Ω and R = 15 Ω. The value of S is
(1) 45 Ω
(2) 7.5 Ω
(3) 15 Ω
(4) 30 Ω

31. A cell of emf 2 V and internal resistance 0.5 Ω is connected to a 3.5 Ω resistor. The current is
(1) 0.5 A
(2) 0.57 A
(3) 4 A
(4) 1 A

32. The magnetic field at the centre of a circular loop of radius R carrying current I is
(1) $\mu_0 I/4R$
(2) $\mu_0 IR/2$
(3) $\mu_0 I/2\pi R$
(4) $\mu_0 I/2R$

33. The cyclotron frequency of a charged particle in a magnetic field B
(1) Is proportional to its kinetic energy
(2) Is independent of speed
(3) Increases with speed
(4) Decreases with speed

34. A galvanometer is converted into an ammeter by connecting
(1) A high resistance in series
(2) A low resistance in series
(3) A low resistance in parallel
(4) A high resistance in parallel

35. The magnetic flux through a coil is $\Phi = 5t^2 + 3t$ weber. The magnitude of induced emf at t = 2 s is
(1) 23 V
(2) 20 V
(3) 13 V
(4) 26 V

36. At resonance in a series LCR circuit
(1) Power factor is zero
(2) Impedance is maximum
(3) Impedance equals R and power factor is 1
(4) Current is minimum

37. An ideal step-up transformer has turns ratio 1 : 10. For 220 V input, the output voltage is
(1) 22 V
(2) 220 V
(3) 22000 V
(4) 2200 V

38. Which of these has the shortest wavelength?
(1) Ultraviolet
(2) Microwaves
(3) X-rays
(4) Gamma rays

39. An object is placed 30 cm from a convex lens of focal length 20 cm. The image is formed at
(1) 12 cm, virtual and erect
(2) 60 cm, real and inverted
(3) 60 cm, virtual and erect
(4) 12 cm, real and inverted

40. In Young's double slit experiment, if the slit separation is halved and the screen distance doubled, the fringe width becomes
(1) Four times
(2) Same
(3) Half
(4) Double

41. Light of photon energy 5 eV falls on a metal of work function 2 eV. The stopping potential is
(1) 3 V
(2) 7 V
(3) 2 V
(4) 5 V

42. The de Broglie wavelength of an electron accelerated through 100 V is about
(1) 12.27 Å
(2) 1.227 Å
(3) 0.1227 Å
(4) 122.7 Å

43. After three half-lives, the fraction of a radioactive sample remaining is
(1) 1/3
(2) 1/8
(3) 1/9
(4) 1/6

44. In Bohr's model of hydrogen, the ratio of the radius of the 2nd orbit to that of the 1st orbit is
(1) 2
(2) 8
(3) 1/2
(4) 4

45. The output of a NAND gate when both inputs are 1 is
(1) 1
(2) 0
(3) Same as input
(4) Undefined

Chemistry

46. The number of moles in 4.4 g of CO_2 is
(1) 0.2
(2) 0.01
(3) 0.1
(4) 1

47. The number of atoms in 18 g of water is
(1) 3N_A
(2) 2N_A
(3) 18N_A
(4) N_A

48. The maximum number of electrons in the shell with n = 3 is
(1) 9
(2) 18
(3) 8
(4) 32

49. Which has the highest first ionisation enthalpy?
(1) C
(2) N
(3) B
(4) O

50. Which has the largest atomic radius?
(1) K
(2) Cs
(3) Na
(4) Li

51. The hybridisation of S in SF_6 is
(1) sp^3
(2) sp^3d
(3) dsp^2
(4) sp^3d^2

52. The shape of XeF_4 is
(1) Square planar
(2) See-saw
(3) Tetrahedral
(4) Square pyramidal

53. Which species has the highest bond order?
(1) O_2^-
(2) O_2
(3) O_2^+
(4) O_2^{2-}

54. The strongest hydrogen bond is present in
(1) HF
(2) NH_3
(3) HCl
(4) H_2O

55. A reaction is spontaneous at all temperatures when
(1) $\Delta H > 0, \Delta S < 0$
(2) $\Delta H < 0, \Delta S > 0$
(3) $\Delta H > 0, \Delta S > 0$
(4) $\Delta H < 0, \Delta S < 0$

56. For $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$, Kp = Kc(RT)^Δn where Δn is
(1) +2
(2) −1
(3) 0
(4) −2

57. The pH of 0.001 M HCl is
(1) 3
(2) 1
(3) 2
(4) 11

58. Increasing pressure on $N_2 + 3H_2 \rightleftharpoons 2NH_3$ at equilibrium
(1) Shifts it forward
(2) Has no effect
(3) Stops the reaction
(4) Shifts it backward

59. The van't Hoff factor of $K_4[Fe(CN)_6]$, assuming complete dissociation, is
(1) 4
(2) 5
(3) 6
(4) 1

60. Given $E°(Cu^{2+}/Cu) = +0.34$ V and $E°(Zn^{2+}/Zn) = -0.76$ V, the E°cell of the Daniell cell is
(1) −1.10 V
(2) 0.76 V
(3) 0.42 V
(4) 1.10 V

61. The charge needed to deposit 1 mol of Al from Al^{3+} is
(1) 3 F
(2) 2 F
(3) 6 F
(4) 1 F

62. A first-order reaction has k = 0.0693 min^{-1}. Its half-life is
(1) 100 min
(2) 1 min
(3) 6.93 min
(4) 10 min

63. For a zero-order reaction, the half-life is
(1) Inversely proportional to initial concentration
(2) Proportional to the square of initial concentration
(3) Directly proportional to initial concentration
(4) Independent of initial concentration

64. Which oxide is amphoteric?
(1) SO_3
(2) Al_2O_3
(3) CaO
(4) Na_2O

65. The most stable +2 oxidation state in group 14 is shown by
(1) Pb
(2) C
(3) Sn
(4) Si

66. Diborane (B_2H_6) contains
(1) Only 2c–2e bonds
(2) Two 3c–2e bonds
(3) Four 3c–2e bonds
(4) A B–B bond

67. The strongest oxidising agent among the halogens is
(1) I_2
(2) F_2
(3) Br_2
(4) Cl_2

68. The strongest acid in aqueous solution is
(1) HCl
(2) HBr
(3) HF
(4) HI

69. The oxidation state of Cr in $K_2Cr_2O_7$ is
(1) +7
(2) +3
(3) +6
(4) +4

70. The spin-only magnetic moment of Fe^{2+} is
(1) 4.90 BM
(2) 3.87 BM
(3) 5.92 BM
(4) 2.83 BM

71. In acidic medium, $KMnO_4$ is reduced to
(1) Mn^{2+}
(2) MnO_4^{2-}
(3) Mn^{3+}
(4) MnO_2

72. Zr and Hf have almost identical atomic radii due to
(1) Lanthanoid contraction
(2) Actinoid contraction
(3) Inert pair effect
(4) Shielding by s-electrons

73. The number of geometrical isomers of $[Co(NH_3)_4Cl_2]^+$ is
(1) 1
(2) 2
(3) 3
(4) 4

74. $[Fe(CN)_6]^{4-}$ is
(1) Diamagnetic, low spin
(2) Paramagnetic, low spin
(3) Paramagnetic, high spin
(4) Tetrahedral

75. The coordination number of Fe in $[Fe(C_2O_4)_3]^{3-}$ is
(1) 4
(2) 3
(3) 6
(4) 9

76. The IUPAC name of $(CH_3)_2CH–CH_2OH$ is
(1) 2-Methylpropan-2-ol
(2) Butan-1-ol
(3) Butan-2-ol
(4) 2-Methylpropan-1-ol

77. The most stable carbocation is
(1) $(CH_3)_3C^+$
(2) $CH_3CH_2^+$
(3) CH_3^+
(4) $(CH_3)_2CH^+$

78. The strongest acid is
(1) CH_3COOH
(2) Cl_3CCOOH
(3) $Cl_2CHCOOH$
(4) $ClCH_2COOH$

79. Propene reacts with HBr (no peroxide) to give mainly
(1) 1-Bromopropane
(2) Propane
(3) 2-Bromopropane
(4) 1,2-Dibromopropane

80. Propene reacts with HBr in the presence of peroxide to give mainly
(1) 2-Bromopropane
(2) Allyl bromide
(3) 1-Bromopropane
(4) Propan-2-ol

81. Which is aromatic?
(1) Cycloheptatriene
(2) Cyclopentadienyl cation
(3) Cyclooctatetraene
(4) Cyclopentadienyl anion

82. The fastest SN1 reaction is shown by
(1) CH_3CH_2Br
(2) $(CH_3)_2CHBr$
(3) $(CH_3)_3CBr$
(4) CH_3Br

83. Phenol treated with $CHCl_3$ and aqueous NaOH, followed by acidification, gives
(1) Salicylaldehyde
(2) Benzoic acid
(3) Benzaldehyde
(4) Salicylic acid

84. Which does NOT undergo aldol condensation?
(1) Benzaldehyde
(2) Acetaldehyde
(3) Acetone
(4) Propanal

85. In the Cannizzaro reaction, HCHO gives
(1) Formic acid only
(2) Methanol and formate ion
(3) Ethanol only
(4) Acetaldehyde

86. The Hinsberg reagent is
(1) Benzenesulphonyl chloride
(2) Benzoyl chloride
(3) Phenyl isocyanide
(4) Acetyl chloride

87. The carbylamine test is given by
(1) Primary amines
(2) Secondary amines
(3) Quaternary ammonium salts
(4) Tertiary amines

88. Which is a non-reducing sugar?
(1) Fructose
(2) Sucrose
(3) Glucose
(4) Maltose

89. The base present in DNA but not in RNA is
(1) Adenine
(2) Cytosine
(3) Guanine
(4) Thymine

90. Deficiency of vitamin C causes
(1) Beriberi
(2) Rickets
(3) Scurvy
(4) Night blindness

Botany

91. Heterocysts in Nostoc are the sites of
(1) Nitrogen fixation
(2) Photosynthesis
(3) Food storage
(4) Asexual reproduction

92. Mycoplasma differs from other bacteria in lacking
(1) DNA
(2) Ribosomes
(3) Cell wall
(4) Plasma membrane

93. Agar is commercially obtained from
(1) Gelidium and Gracilaria
(2) Laminaria and Sargassum
(3) Chlorella and Spirulina
(4) Fucus and Dictyota

94. Which group is called the 'amphibians of the plant kingdom'?
(1) Pteridophytes
(2) Algae
(3) Bryophytes
(4) Gymnosperms

95. The coralloid roots of Cycas are associated with
(1) Mycorrhizal fungi
(2) Cyanobacteria
(3) Frankia
(4) Rhizobium

96. Vexillary aestivation is characteristic of
(1) Liliaceae
(2) Solanaceae
(3) Brassicaceae
(4) Fabaceae

97. The aestivation in China rose (Hibiscus) is
(1) Twisted
(2) Imbricate
(3) Vexillary
(4) Valvate

98. The placentation in mustard is
(1) Basal
(2) Free central
(3) Parietal
(4) Axile

99. Scattered, closed vascular bundles are found in
(1) Monocot stem
(2) Monocot root
(3) Dicot root
(4) Dicot stem

100. Casparian strips are found in the
(1) Endodermis
(2) Epidermis
(3) Pericycle
(4) Cortex

101. Annual rings are formed due to the activity of
(1) Vascular cambium
(2) Intercalary meristem
(3) Apical meristem
(4) Cork cambium

102. 70S ribosomes are found in prokaryotes and in
(1) Golgi bodies
(2) Mitochondria and chloroplasts
(3) Nucleus
(4) Lysosomes

103. The fluid mosaic model of the plasma membrane was given by
(1) Watson and Crick
(2) Robertson
(3) Singer and Nicolson
(4) Schleiden and Schwann

104. The synaptonemal complex forms during
(1) Pachytene
(2) Zygotene
(3) Leptotene
(4) Diakinesis

105. Crossing over occurs during
(1) Leptotene
(2) Pachytene
(3) Zygotene
(4) Diplotene

106. Inhibition of succinic dehydrogenase by malonate is
(1) Non-competitive
(2) Competitive
(3) Allosteric
(4) Irreversible

107. The water potential of pure water at atmospheric pressure is
(1) Negative
(2) Positive
(3) Infinite
(4) Zero

108. Plasmolysis occurs when a cell is placed in a
(1) Isotonic solution
(2) Hypotonic solution
(3) Distilled water
(4) Hypertonic solution

109. Leghaemoglobin in root nodules
(1) Protects nitrogenase from oxygen
(2) Fixes nitrogen
(3) Stores nitrogen
(4) Transports nitrate

110. In C_4 plants, the primary CO_2 acceptor is
(1) PEP
(2) OAA
(3) RuBP
(4) PGA

111. The reaction centre of Photosystem I is
(1) P700
(2) P690
(3) P870
(4) P680

112. To make one glucose molecule, the Calvin cycle needs
(1) 30 ATP and 12 NADPH
(2) 6 ATP and 6 NADPH
(3) 18 ATP and 12 NADPH
(4) 12 ATP and 18 NADPH

113. According to NCERT, the net ATP gain from complete aerobic oxidation of one glucose molecule is
(1) 8
(2) 2
(3) 36
(4) 38

114. The respiratory quotient of fats is
(1) More than 1
(2) 1
(3) Less than 1
(4) Infinity

115. Bolting in rosette plants is caused by
(1) Gibberellin
(2) Auxin
(3) Abscisic acid
(4) Cytokinin

116. Apical dominance is caused by
(1) Cytokinin
(2) Auxin
(3) Ethylene
(4) ABA

117. The hormone that promotes fruit ripening is
(1) Ethylene
(2) Gibberellin
(3) Auxin
(4) Cytokinin

118. Perisperm is
(1) Remnant of the secondary nucleus
(2) Persistent nucellus
(3) Remnant of endosperm
(4) The seed coat

119. In angiosperms, the endosperm is typically
(1) Diploid
(2) Haploid
(3) Triploid
(4) Tetraploid

120. Cleistogamous flowers are
(1) Pollinated by insects
(2) Invariably autogamous
(3) Always cross-pollinated
(4) Unisexual

121. Apomixis is
(1) Development of an embryo from a zygote
(2) Fruit formation without fertilisation
(3) Germination inside the fruit
(4) Seed formation without fertilisation

122. The phenotypic ratio of a Mendelian dihybrid cross in F_2 is
(1) 9 : 3 : 3 : 1
(2) 3 : 1
(3) 1 : 2 : 1
(4) 1 : 1 : 1 : 1

123. A test cross is a cross between
(1) F_1 and the dominant parent
(2) Two F_1 plants
(3) F_1 and the recessive parent
(4) Two pure lines

124. The F_2 phenotypic ratio for flower colour in snapdragon (incomplete dominance) is
(1) 3 : 1
(2) 9 : 7
(3) 1 : 2 : 1
(4) 1 : 1

125. Linkage was first demonstrated experimentally by Morgan in
(1) Drosophila
(2) Maize
(3) Pea
(4) Neurospora

126. Semi-conservative DNA replication was proved by
(1) Griffith
(2) Avery, MacLeod and McCarty
(3) Hershey and Chase
(4) Meselson and Stahl

127. Okazaki fragments are formed on the
(1) Both strands equally
(2) Template of mRNA
(3) Leading strand
(4) Lagging strand

128. The inducer of the lac operon is
(1) Glucose
(2) Lactose (allolactose)
(3) Galactose
(4) Repressor

129. The codon AUG codes for
(1) Valine
(2) Methionine
(3) Tryptophan
(4) Stop

130. An inverted pyramid of biomass is seen in
(1) Grassland
(2) Sea
(3) Forest
(4) Desert

131. The 10 percent law of energy transfer was given by
(1) Lindeman
(2) Tansley
(3) Odum
(4) Elton

132. Species diversity on Earth is highest in the
(1) Tropical rain forests (e.g. Amazon)
(2) Taiga
(3) Tundra
(4) Temperate forests

133. An example of ex-situ conservation is a
(1) Biosphere reserve
(2) National park
(3) Seed bank
(4) Sacred grove

134. An interaction where one species benefits and the other is unaffected is
(1) Amensalism
(2) Parasitism
(3) Commensalism
(4) Mutualism

135. Logistic population growth is described by
(1) $N = N_0 e^{rt}$
(2) $dN/dt = rN(K - N)/K$
(3) $dN/dt = rN$
(4) $dN/dt = K$

Zoology

136. Which is a cyclostome?
(1) Pristis
(2) Torpedo
(3) Petromyzon
(4) Scoliodon

137. The water vascular system is characteristic of
(1) Echinodermata
(2) Mollusca
(3) Porifera
(4) Annelida

138. Flame cells are excretory structures of
(1) Aschelminthes
(2) Annelida
(3) Arthropoda
(4) Platyhelminthes

139. Metagenesis is seen in
(1) Ascaris
(2) Taenia
(3) Obelia
(4) Hydra

140. In cockroach, excretion is carried out by
(1) Flame cells
(2) Green glands
(3) Malpighian tubules
(4) Nephridia

141. The epithelium of the proximal convoluted tubule is
(1) Simple squamous
(2) Stratified squamous
(3) Cuboidal with brush border
(4) Ciliated columnar

142. A ligament connects
(1) Muscle to bone
(2) Skin to muscle
(3) Bone to bone
(4) Muscle to muscle

143. Pepsinogen is converted to pepsin by
(1) Enterokinase
(2) Trypsin
(3) Bile
(4) HCl

144. Trypsinogen is activated by
(1) Enterokinase
(2) Pepsin
(3) Lipase
(4) HCl

145. Bile helps in
(1) Absorbing glucose
(2) Emulsifying fats
(3) Digesting proteins
(4) Digesting starch

146. The tidal volume in a healthy adult is about
(1) 2500 mL
(2) 1200 mL
(3) 500 mL
(4) 3000 mL

147. The oxygen dissociation curve shifts to the right with
(1) High pCO_2, high H^+ and high temperature
(2) Low temperature
(3) Low pCO_2
(4) High pH

148. The universal blood donor group is
(1) AB^-
(2) O^-
(3) AB^+
(4) O^+

149. The duration of one cardiac cycle at 72 beats per minute is
(1) 0.8 s
(2) 0.08 s
(3) 0.4 s
(4) 1.2 s

150. The pacemaker of the human heart is the
(1) SA node
(2) AV node
(3) Purkinje fibres
(4) Bundle of His

151. ADH increases water reabsorption in the
(1) Descending limb of Henle's loop
(2) DCT and collecting duct
(3) Bowman's capsule
(4) PCT

152. The counter-current mechanism involves
(1) DCT and collecting duct
(2) PCT and glomerulus
(3) Bowman's capsule and PCT
(4) Henle's loop and vasa recta

153. Gout is caused by the accumulation of
(1) Calcium oxalate in kidneys
(2) Ammonia
(3) Uric acid crystals in joints
(4) Urea

154. During muscle contraction, the band whose length stays constant is the
(1) H zone
(2) Z line
(3) A band
(4) I band

155. During muscle contraction, Ca^{2+} binds to
(1) Actin
(2) Troponin
(3) Myosin
(4) Tropomyosin

156. The number of cervical vertebrae in humans is
(1) 5
(2) 12
(3) 4
(4) 7

157. At rest, the axonal membrane is more permeable to
(1) Cl^-
(2) Na^+
(3) K^+
(4) Ca^{2+}

158. Saltatory conduction occurs in
(1) Synapses
(2) Myelinated fibres
(3) Non-myelinated fibres
(4) Dendrites only

159. Insulin is secreted by the
(1) δ-cells
(2) α-cells
(3) β-cells
(4) PP cells

160. Hypothyroidism in adults causes
(1) Cretinism
(2) Myxoedema
(3) Exophthalmic goitre
(4) Acromegaly

161. Atrial natriuretic factor
(1) Raises blood pressure
(2) Increases Na^+ reabsorption
(3) Lowers blood pressure
(4) Raises blood glucose

162. Sertoli cells
(1) Form the acrosome
(2) Produce semen
(3) Secrete testosterone
(4) Nourish developing sperm

163. Testosterone is secreted by
(1) Sertoli cells
(2) Leydig cells
(3) Prostate
(4) Spermatogonia

164. Ovulation is triggered by a surge of
(1) Progesterone
(2) Oxytocin
(3) LH
(4) FSH

165. The corpus luteum mainly secretes
(1) LH
(2) Oestrogen
(3) Progesterone
(4) hCG

166. In humans, implantation occurs at the
(1) Zygote stage
(2) Gastrula stage
(3) Blastocyst stage
(4) Morula stage

167. Copper-releasing IUDs act by
(1) Suppressing sperm motility and fertilising capacity
(2) Killing the ovum
(3) Stopping menstruation
(4) Blocking ovulation

168. In ZIFT, the embryo is transferred into the fallopian tube up to the
(1) Blastocyst stage
(2) Gastrula stage
(3) 8-blastomere stage
(4) 32-cell stage

169. Amniocentesis is misused for
(1) Checking foetal heart rate
(2) Detecting chromosomal disorders
(3) Detecting metabolic disorders
(4) Foetal sex determination

170. Homologous organs indicate
(1) No evolution
(2) Parallel evolution
(3) Divergent evolution
(4) Convergent evolution

171. In Miller's experiment, the gases used were
(1) CH_4, O_2, N_2
(2) CO_2, O_2, N_2
(3) NH_3, O_2, CO_2
(4) CH_4, NH_3, H_2 and water vapour

172. In a Hardy–Weinberg population, the frequency of homozygous recessives is 0.16. The frequency of heterozygotes is
(1) 0.36
(2) 0.16
(3) 0.48
(4) 0.24

173. The infective stage of Plasmodium for humans is the
(1) Merozoite
(2) Trophozoite
(3) Sporozoite
(4) Gametocyte

174. Typhoid is confirmed by the
(1) Schick test
(2) Widal test
(3) ELISA test
(4) Mantoux test

175. The antibody abundant in colostrum is
(1) IgG
(2) IgE
(3) IgA
(4) IgM

176. HIV mainly destroys
(1) Platelets
(2) B-lymphocytes
(3) Helper T-lymphocytes
(4) Red blood cells

177. EcoRI recognises the palindromic sequence
(1) GAATTC
(2) CTGCAG
(3) GGATCC
(4) AAGCTT

178. The thermostable DNA polymerase used in PCR comes from
(1) Agrobacterium
(2) Escherichia coli
(3) Bacillus thuringiensis
(4) Thermus aquaticus

179. RNA interference was used to protect tobacco from
(1) Bollworm
(2) Meloidogyne incognita
(3) Corn borer
(4) Aphids

180. Bt toxin proteins are coded by
(1) nif genes
(2) hox genes
(3) lac genes
(4) cry genes

Answer key

1 → 3	**2** → 2	**3** → 3	**4** → 2	**5** → 2	**6** → 2	**7** → 3	**8** → 3	**9** → 4	**10** → 2
11 → 3	**12** → 2	**13** → 1	**14** → 3	**15** → 2	**16** → 4	**17** → 4	**18** → 2	**19** → 4	**20** → 4
21 → 1	**22** → 3	**23** → 2	**24** → 1	**25** → 2	**26** → 4	**27** → 1	**28** → 1	**29** → 2	**30** → 4
31 → 1	**32** → 4	**33** → 2	**34** → 3	**35** → 1	**36** → 3	**37** → 4	**38** → 4	**39** → 2	**40** → 1
41 → 1	**42** → 2	**43** → 2	**44** → 4	**45** → 2	**46** → 3	**47** → 1	**48** → 2	**49** → 2	**50** → 2
51 → 4	**52** → 1	**53** → 3	**54** → 1	**55** → 2	**56** → 4	**57** → 1	**58** → 1	**59** → 2	**60** → 4
61 → 1	**62** → 4	**63** → 3	**64** → 2	**65** → 1	**66** → 2	**67** → 2	**68** → 4	**69** → 3	**70** → 1
71 → 1	**72** → 1	**73** → 2	**74** → 1	**75** → 3	**76** → 4	**77** → 1	**78** → 2	**79** → 3	**80** → 3
81 → 4	**82** → 3	**83** → 1	**84** → 1	**85** → 2	**86** → 1	**87** → 1	**88** → 2	**89** → 4	**90** → 3
91 → 1	**92** → 3	**93** → 1	**94** → 3	**95** → 2	**96** → 4	**97** → 1	**98** → 3	**99** → 1	**100** → 1
101 → 1	**102** → 2	**103** → 3	**104** → 2	**105** → 2	**106** → 2	**107** → 4	**108** → 4	**109** → 1	**110** → 1
111 → 1	**112** → 3	**113** → 4	**114** → 3	**115** → 1	**116** → 2	**117** → 1	**118** → 2	**119** → 3	**120** → 2
121 → 4	**122** → 1	**123** → 3	**124** → 3	**125** → 1	**126** → 4	**127** → 4	**128** → 2	**129** → 2	**130** → 2
131 → 1	**132** → 1	**133** → 3	**134** → 3	**135** → 2	**136** → 3	**137** → 1	**138** → 4	**139** → 3	**140** → 3
141 → 3	**142** → 3	**143** → 4	**144** → 1	**145** → 2	**146** → 3	**147** → 1	**148** → 2	**149** → 1	**150** → 1
151 → 2	**152** → 4	**153** → 3	**154** → 3	**155** → 2	**156** → 4	**157** → 3	**158** → 2	**159** → 3	**160** → 2
161 → 3	**162** → 4	**163** → 2	**164** → 3	**165** → 3	**166** → 3	**167** → 1	**168** → 3	**169** → 4	**170** → 3
171 → 4	**172** → 3	**173** → 3	**174** → 2	**175** → 3	**176** → 3	**177** → 1	**178** → 4	**179** → 2	**180** → 4

Explanations

1. (3) h = E/ν has dimensions $[ML^2T^{-1}]$, the same as angular momentum (mvr). *[Units & Dimensions]*

2. (2) H = $u^2/2g$ = 400/20 = 20 m. *[Kinematics]*

3. (3) R = $u^2 \sin 2\theta/g$; $\sin 60° = \sin 120°$, so the ranges are equal for complementary angles. *[Kinematics]*

4. (2) a = F/m = 20/5 = 4 m s^{-2}. *[Laws of Motion]*

5. (2) At the angle of repose, $\tan\theta = \mu s = 0.5$. *[Laws of Motion]*

6. (2) p = $\sqrt{(2mK)}$ = $\sqrt{(2 \times 2 \times 100)}$ = 20 kg m s^{-1}. *[Work, Energy & Power]*

7. (3) W = ½kx^2 = ½ × 200 × $(0.1)^2$ = 1 J. *[Work, Energy & Power]*

8. (3) In a 1-D elastic collision between equal masses, the velocities are exchanged. *[Work, Energy & Power]*

9. (4) Parallel axis theorem: (2/5)MR^2 + MR^2 = (7/5)MR^2. *[Rotational Motion]*

10. (2) K_rot = ¼Mv^2, K_total = ¾Mv^2, so fraction = 1/3. *[Rotational Motion]*

11. (3) Angular momentum is conserved: Iω = 2I·ω′ ⇒ ω′ = ω/2. *[Rotational Motion]*

12. (2) v_e = $\sqrt{(2GM/R)} \propto R\sqrt{\rho}$, so doubling R doubles v_e. *[Gravitation]*

13. (1) v_e = $\sqrt{(2gR)}$, v_o = $\sqrt{(gR)}$, ratio = $\sqrt{2}$. *[Gravitation]*

14. (3) ΔL = FL/(AY); L doubles, A becomes 4 times, so ΔL′ = ΔL/2. *[Mechanical Properties of Solids]*

15. (2) $v_t = 2r^2(\rho - \sigma)g/9\eta \propto r^2$. *[Mechanical Properties of Fluids]*

16. (4) Continuity: $A_1v_1 = A_2v_2 \Rightarrow v_2 = 4 \times 2 / 1$ = 8 m s^{-1}. *[Mechanical Properties of Fluids]*

17. (4) A soap bubble has two surfaces, so ΔP = 4T/r. *[Mechanical Properties of Fluids]*

18. (2) $\eta = 1 - T_2/T_1 = 1 - 300/500 = 0.4$. *[Thermodynamics]*

19. (4) v_rms ∝ √T; T goes from 300 K to 600 K, so v_rms × √2. *[Kinetic Theory]*

20. (4) f = 5 ⇒ γ = 1 + 2/f = 7/5. *[Kinetic Theory]*

21. (1) ω = π ⇒ T = 2π/ω = 2 s. *[Oscillations]*

22. (3) T ∝ √L; √1.21 = 1.1 ⇒ 10% increase. *[Oscillations]*

23. (2) f = 50 Hz, 1/λ = 0.5 ⇒ λ = 2 m, v = fλ = 100 m s^{-1}. *[Waves]*

24. (1) A closed pipe has only odd harmonics: the first overtone is the 3rd harmonic, 300 Hz. *[Waves]*

25. (2) $F \propto q_1q_2/r^2$: $(2 \times 2)/(2^2) = 1$. *[Electrostatics]*

26. (4) Forces on +q and −q are equal and opposite (zero net force) but form a couple (τ = pE sinθ). *[Electrostatics]*

27. (1) 1/C = 3/6 ⇒ C = 2 μF. *[Capacitance]*

28. (1) U = ½CV^2 = ½ × 10^{-5} × 10^4 = 0.05 J. *[Capacitance]*

29. (2) Volume is constant: L doubles, A halves ⇒ R′ = 4R. *[Current Electricity]*

30. (4) P/Q = R/S ⇒ S = 15 × 20/10 = 30 Ω. *[Current Electricity]*

31. (1) I = E/(R + r) = 2/4 = 0.5 A. *[Current Electricity]*

32. (4) $B = \mu_0 I/2R$ at the centre of a circular loop. *[Moving Charges & Magnetism]*

33. (2) $f = qB/2\pi m$ does not depend on speed (non-relativistic). *[Moving Charges & Magnetism]*

34. (3) A small shunt in parallel diverts most of the current. *[Moving Charges & Magnetism]*

35. (1) $|e| = d\Phi/dt = 10t + 3 = 23$ V. *[Electromagnetic Induction]*

36. (3) $X_L = X_C \Rightarrow Z = R$, $\cos\varphi = 1$, current is maximum. *[Alternating Current]*

37. (4) $V_s/V_p = N_s/N_p = 10$. *[Alternating Current]*

38. (4) Gamma rays lie at the highest-frequency end of the EM spectrum. *[Electromagnetic Waves]*

39. (2) $1/v = 1/f + 1/u = 1/20 - 1/30 = 1/60 \Rightarrow v = +60$ cm, real and inverted. *[Ray Optics]*

40. (1) $\beta = \lambda D/d$: D × 2 and d ÷ 2 ⇒ β × 4. *[Wave Optics]*

41. (1) $eV_0 = h\nu - \varphi = 3$ eV ⇒ $V_0 = 3$ V. *[Dual Nature]*

42. (2) $\lambda = 12.27/\sqrt{V}$ Å = 1.227 Å. *[Dual Nature]*

43. (2) $(1/2)^3 = 1/8$. *[Atoms & Nuclei]*

44. (4) $r \propto n^2 \Rightarrow r_2/r_1 = 4$. *[Atoms & Nuclei]*

45. (2) NAND = NOT(AND); AND(1,1) = 1 ⇒ NAND = 0. *[Semiconductors]*

46. (3) Molar mass of CO_2 = 44 g mol^{-1}; 4.4/44 = 0.1. *[Some Basic Concepts]*

47. (1) 18 g = 1 mol H_2O, and each molecule has 3 atoms. *[Some Basic Concepts]*

48. (2) $2n^2 = 18$. *[Atomic Structure]*

49. (2) N has a stable half-filled $2p^3$ configuration, so its IE_1 is higher than O. *[Classification of Elements]*

50. (2) Atomic size increases down a group. *[Classification of Elements]*

51. (4) Six bond pairs, no lone pair ⇒ sp^3d^2, octahedral. *[Chemical Bonding]*

52. (1) 4 bond pairs + 2 lone pairs (sp^3d^2) ⇒ square planar. *[Chemical Bonding]*

53. (3) Bond orders: O_2^+ 2.5, O_2 2, O_2^- 1.5, O_2^{2-} 1. *[Chemical Bonding]*

54. (1) F is the most electronegative, so the F–H···F bond is strongest. *[Chemical Bonding]*

55. (2) $\Delta G = \Delta H - T\Delta S$ is negative at every T only if $\Delta H < 0$ and $\Delta S > 0$. *[Thermodynamics]*

56. (4) $\Delta n = 2 - 4 = -2$. *[Equilibrium]*

57. (1) $[H^+] = 10^{-3}$ M ⇒ pH = 3. *[Equilibrium]*

58. (1) Le Chatelier: higher pressure favours the side with fewer gas moles. *[Equilibrium]*

59. (2) $4K^+ + [Fe(CN)_6]^{4-} = 5$ ions. *[Solutions]*

60. (4) $E°_{cell} = E°_{cathode} - E°_{anode} = 0.34 - (-0.76) = 1.10$ V. *[Electrochemistry]*

61. (1) $Al^{3+} + 3e^- \rightarrow Al$. *[Electrochemistry]*

62. (4) $t_{½} = 0.693/k = 10$ min. *[Chemical Kinetics]*

63. (3) $t_{½} = [A]_0/2k$. *[Chemical Kinetics]*

64. (2) Al_2O_3 reacts with both acids and bases. *[p-Block]*

65. (1) The inert pair effect increases down the group. *[p-Block]*

66. (2) Two bridging B–H–B 'banana' bonds are 3c–2e bonds. *[p-Block]*

67. (2) F_2 has the highest standard reduction potential. *[p-Block]*

68. (4) The H–I bond is the weakest, so HI ionises most readily. *[p-Block]*

69. (3) $2(+1) + 2x + 7(-2) = 0 \Rightarrow x = +6$. *[d- and f-Block]*

70. (1) d^6 high spin has 4 unpaired electrons: $\sqrt{(4\times6)} = 4.90$ BM. *[d- and f-Block]*

71. (1) $MnO_4^- + 8H^+ + 5e^- \rightarrow Mn^{2+} + 4H_2O$. *[d- and f-Block]*

72. (1) Poor shielding by 4f electrons shrinks the 5d elements. *[d- and f-Block]*

73. (2) MA_4B_2 octahedral: cis and trans. *[Coordination Compounds]*

74. (1) CN^- is a strong-field ligand: Fe^{2+} d^6 becomes t_2g^6 with no unpaired electrons. *[Coordination Compounds]*

75. (3) Oxalate is bidentate: 3 × 2 = 6. *[Coordination Compounds]*

76. (4) Longest chain with –OH: 3 carbons, methyl at C-2. *[Organic: Basic Principles]*

77. (1) Tertiary carbocations are stabilised most by hyperconjugation and +I effect. *[Organic: Basic Principles]*

78. (2) Three –I chlorines stabilise the carboxylate the most. *[Organic: Basic Principles]*

79. (3) Markovnikov addition via the more stable 2° carbocation. *[Hydrocarbons]*

80. (3) Peroxide (Kharasch) effect gives anti-Markovnikov addition. *[Hydrocarbons]*

81. (4) The anion is planar, cyclic, conjugated, with 6π electrons ($4n + 2$, $n = 1$). *[Hydrocarbons]*

82. (3) SN1 goes through a carbocation; a 3° carbocation is most stable. *[Haloalkanes]*

83. (1) The Reimer–Tiemann reaction gives ortho-hydroxybenzaldehyde. *[Alcohols & Phenols]*

84. (1) Benzaldehyde has no α-hydrogen. *[Aldehydes & Ketones]*

85. (2) Disproportionation: one HCHO is reduced to CH_3OH and another is oxidised to $HCOO^-$. *[Aldehydes & Ketones]*

86. (1) $C_6H_5SO_2Cl$ distinguishes 1°, 2° and 3° amines. *[Amines]*

87. (1) Only 1° amines + $CHCl_3$/KOH give foul-smelling isocyanides. *[Amines]*

88. (2) In sucrose the glycosidic link joins both anomeric carbons. *[Biomolecules]*

89. (4) RNA has uracil in place of thymine. *[Biomolecules]*

90. (3) Vitamin C (ascorbic acid) is needed for collagen synthesis. *[Biomolecules]*

91. (1) Heterocysts provide an anaerobic site for nitrogenase. *[Biological Classification]*

92. (3) Mycoplasma has no cell wall and is pleomorphic. *[Biological Classification]*

93. (1) Agar comes from the red algae Gelidium and Gracilaria. *[Plant Kingdom]*

94. (3) Bryophytes live on land but need water for fertilisation. *[Plant Kingdom]*

95. (2) N_2-fixing cyanobacteria (Anabaena) live in coralloid roots. *[Plant Kingdom]*

96. (4) Pea flowers have a standard, two wings and a keel. *[Morphology of Flowering Plants]*

97. (1) Each petal overlaps the next one on one side. *[Morphology of Flowering Plants]*

98. (3) Ovules are on the ovary wall; the ovary becomes two-chambered by a false septum. *[Morphology of Flowering Plants]*

99. (1) Monocot stems have scattered vascular bundles without cambium. *[Anatomy of Flowering Plants]*

100. (1) Suberin bands on the radial walls of endodermal cells. *[Anatomy of Flowering Plants]*

101. (1) Seasonal differences in vascular cambium activity produce spring and autumn wood. *[Anatomy of Flowering Plants]*

102. (2) These semi-autonomous organelles have their own 70S ribosomes. *[Cell: Unit of Life]*

103. (3) Singer and Nicolson (1972). *[Cell: Unit of Life]*

104. (2) Homologous chromosomes pair (synapsis) in zygotene. *[Cell Cycle]*

105. (2) Recombination nodules form and crossing over occurs in pachytene. *[Cell Cycle]*

106. (2) Malonate resembles succinate and competes for the active site. *[Biomolecules]*

107. (4) By convention, Ψw of pure water = 0. *[Transport in Plants]*

108. (4) Water leaves the cell by exosmosis. *[Transport in Plants]*

109. (1) It is an oxygen scavenger that keeps nitrogenase active. *[Mineral Nutrition]*

110. (1) PEP carboxylase fixes CO_2 in mesophyll cells. *[Photosynthesis]*

111. (1) PS I absorbs maximally at 700 nm. *[Photosynthesis]*

112. (3) Six turns of the cycle, each using 3 ATP and 2 NADPH. *[Photosynthesis]*

113. (4) NCERT uses the theoretical yield of 38 ATP. *[Respiration]*

114. (3) Fats need more O_2 than the CO_2 they release (RQ ≈ 0.7). *[Respiration]*

115. (1) GAs promote internode elongation before flowering. *[Plant Growth]*

116. (2) Auxin from the apical bud suppresses lateral buds. *[Plant Growth]*

117. (1) Ethylene is the ripening hormone. *[Plant Growth]*

118. (2) Persistent nucellus in seeds such as black pepper and beet. *[Sexual Reproduction in Plants]*

119. (3) It forms from triple fusion (2 polar nuclei + 1 male gamete). *[Sexual Reproduction in Plants]*

120. (2) They never open, so only self-pollination is possible. *[Sexual Reproduction in Plants]*

121. (4) Apomixis is asexual reproduction that mimics sexual reproduction. *[Sexual Reproduction in Plants]*

122. (1) Independent assortment of two gene pairs. *[Principles of Inheritance]*

123. (3) Crossing with a homozygous recessive reveals the genotype. *[Principles of Inheritance]*

124. (3) Phenotypic and genotypic ratios are the same: red : pink : white. *[Principles of Inheritance]*

125. (1) Morgan's dihybrid crosses in Drosophila. *[Principles of Inheritance]*

126. (4) They used $^{15}N/^{14}N$ and CsCl density gradient centrifugation. *[Molecular Basis of Inheritance]*

127. (4) Discontinuous synthesis happens on the lagging strand (3′→5′ template). *[Molecular Basis of Inheritance]*

128. (2) The inducer binds the repressor and inactivates it. *[Molecular Basis of Inheritance]*

129. (2) AUG is the start codon (methionine). *[Molecular Basis of Inheritance]*

130. (2) In the sea, phytoplankton biomass is less than zooplankton biomass at a given time. *[Ecosystem]*

131. (1) Lindeman (1942). *[Ecosystem]*

132. (1) Latitudinal gradient: diversity is highest near the equator. *[Biodiversity]*

133. (3) Ex-situ means conservation outside the natural habitat. *[Biodiversity]*

134. (3) For example, an orchid growing on a mango branch. *[Organisms & Populations]*

135. (2) Verhulst–Pearl logistic equation. *[Organisms & Populations]*

136. (3) Lampreys (Petromyzon) are jawless cyclostomes. *[Animal Kingdom]*

137. (1) It is used for locomotion, food capture and respiration. *[Animal Kingdom]*

138. (4) Flatworms use protonephridia with flame cells. *[Animal Kingdom]*

139. (3) Obelia alternates between polyp and medusa. *[Animal Kingdom]*

140. (3) Malpighian tubules excrete uric acid. *[Structural Organisation]*

141. (3) Microvilli increase surface area for reabsorption. *[Structural Organisation]*

142. (3) Tendons join muscle to bone; ligaments join bone to bone. *[Structural Organisation]*

143. (4) Gastric HCl activates pepsinogen. *[Digestion]*

144. (1) Enterokinase from the intestinal mucosa activates trypsinogen. *[Digestion]*

145. (2) Bile salts break fat globules into micelles. *[Digestion]*

146. (3) About 500 mL per normal breath. *[Breathing & Exchange of Gases]*

147. (1) Bohr effect: O_2 unloads more easily in active tissues. *[Breathing & Exchange of Gases]*

148. (2) O^- has no A, B or Rh antigens. *[Body Fluids & Circulation]*

149. (1) 60/72 ≈ 0.8 s. *[Body Fluids & Circulation]*

150. (1) The SA node produces the most action potentials per minute. *[Body Fluids & Circulation]*

151. (2) Vasopressin makes the distal nephron permeable to water. *[Excretion]*

152. (4) It builds the medullary osmotic gradient. *[Excretion]*

153. (3) Uric acid crystals deposit in joints. *[Excretion]*

154. (3) The A band (thick filament length) is unchanged. *[Locomotion & Movement]*

155. (2) Ca^{2+} binds troponin, exposing the myosin-binding sites on actin. *[Locomotion & Movement]*

156. (4) Almost all mammals have 7 cervical vertebrae. *[Locomotion & Movement]*

157. (3) The resting membrane is far more permeable to K^+ than Na^+. *[Neural Control]*

158. (2) Impulses jump from one node of Ranvier to the next. *[Neural Control]*

159. (3) β-cells of the islets of Langerhans. *[Chemical Coordination]*

160. (2) Myxoedema; cretinism is the childhood form. *[Chemical Coordination]*

161. (3) ANF causes vasodilation and natriuresis. *[Chemical Coordination]*

162. (4) They are the nurse cells of the seminiferous tubules. *[Human Reproduction]*

163. (2) Leydig (interstitial) cells secrete androgens. *[Human Reproduction]*

164. (3) The mid-cycle LH surge causes rupture of the Graafian follicle. *[Human Reproduction]*

165. (3) Progesterone maintains the endometrium. *[Human Reproduction]*

166. (3) The blastocyst embeds through its trophoblast. *[Human Reproduction]*

167. (1) Cu ions suppress sperm motility. *[Reproductive Health]*

168. (3) Zygote intra-fallopian transfer: up to 8 blastomeres. *[Reproductive Health]*

169. (4) This is why it is legally banned for sex determination. *[Reproductive Health]*

170. (3) Same basic structure, different functions. *[Evolution]*

171. (4) He simulated a reducing primitive atmosphere. *[Evolution]*

172. (3) q = 0.4, p = 0.6, 2pq = 0.48. *[Evolution]*

173. (3) Sporozoites enter through the bite of an infected female Anopheles. *[Human Health & Disease]*

174. (2) The Widal test detects Salmonella typhi. *[Human Health & Disease]*

175. (3) IgA gives the newborn passive immunity. *[Human Health & Disease]*

176. (3) It infects $CD4^+$ helper T cells. *[Human Health & Disease]*

177. (1) EcoRI cuts 5′-G↓AATTC-3′. *[Biotechnology: Principles]*

178. (4) Taq polymerase. *[Biotechnology: Principles]*

179. (2) Silencing the nematode's mRNA protects tobacco roots. *[Biotechnology: Applications]*

180. (4) cry genes of Bacillus thuringiensis. *[Biotechnology: Applications]*